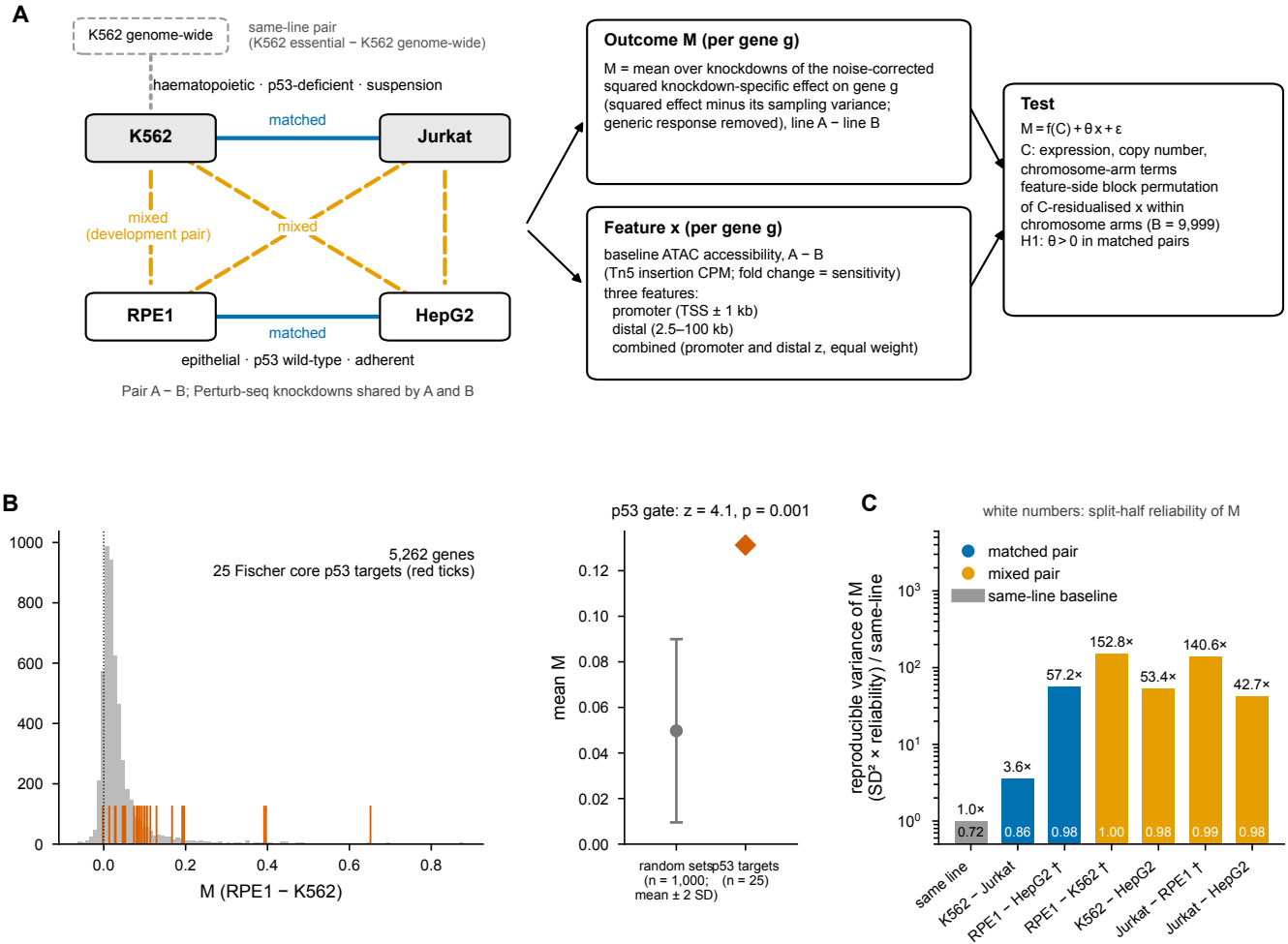

A
K562 genome-wide
same-line pair
(K562 essential − K562 genome-wide)
haematopoietic · p53-deficient · suspension
K562
matched
Jurkat
mixed
(development pair)
mixed
RPE1
matched
HepG2
epithelial · p53 wild-type · adherent
Pair A − B; Perturb-seq knockdowns shared by A and B
Outcome M (per gene g)
M = mean over knockdowns of the noise-corrected squared knockdown-specific effect on gene g (squared effect minus its sampling variance; generic response removed), line A − line B
Feature x (per gene g)
baseline ATAC accessibility, A − B
(Tn5 insertion CPM; fold change = sensitivity)
three features:
promoter (TSS ± 1 kb)
distal (2.5–100 kb)
combined (promoter and distal z, equal weight)
Test
M = f(C) + θ x + ε
C: expression, copy number, chromosome-arm terms
feature-side block permutation
of C-residualised x within chromosome arms (B = 9,999)
H1: θ > 0 in matched pairs
B
5,262 genes
25 Fischer core p53 targets (red ticks)
genes
M (RPE1 − K562)
p53 gate: z = 4.1, p = 0.001
mean M
random sets
(n = 1,000;
mean ± 2 SD)
p53 targets
(n = 25)
C
white numbers: split-half reliability of M
matched pair
mixed pair
same-line baseline
reproducible variance of M
(SD² × reliability) / same-line
1.0×
3.6×
57.2×
152.8×
53.4×
140.6×
42.7×
0.72
0.86
0.98
1.00
0.98
0.99
0.98
same line
K562 − Jurkat
RPE1 − HepG2 †
RPE1 − K562 †
K562 − HepG2
Jurkat − RPE1 †
Jurkat − HepG2